Mathematical Foundations of the Unified Substrate Theory (UST)

1. Definitions

Definition 1 (Substrate Field).

Let

$$\phi: \mathbb{R}^3 \times \mathbb{R} \to \mathbb{R}$$

be the **substrate field**, representing the local activity level of the physical substrate.

Definition 2 (Activity Density).

Define the **activity density**

$$a(\boldsymbol{x}, t) = f\big(\phi(\boldsymbol{x}, t)\big),$$

where $f$is a monotonic mapping. For simplicity, take $f(\phi) = \phi$.

Definition 3 (Energy Density).

Associate an energy density

$$\rho(\boldsymbol{x}, t) = g\big(a(\boldsymbol{x}, t)\big),$$

with $g$monotonic. In the simplest model, $g(a) = a$.

Definition 4 (Universal Baseline Level — UBL).

Let $\rho_P$be the **Planck energy density**. Define the **local UBL**

$$u(\boldsymbol{x}, t) = \frac{\rho(\boldsymbol{x}, t)}{\rho_P}.$$

Definition 5 (Noise Band).

Let $N(\boldsymbol{x}, t)$be the effective number of independent degrees of freedom in a region. Define fractional noise

$$\eta(\boldsymbol{x}, t) = \frac{1}{\sqrt{N(\boldsymbol{x}, t)}}.$$

Definition 6 (Tension Density).

Define substrate tension

$$\mathcal{T}(\boldsymbol{x}, t) = \frac{\mu}{2} \mid \nabla u(\boldsymbol{x}, t) \mid^2,$$

with $\mu > 0$a tension coefficient.

Definition 7 (Damping).

Define damping toward a baseline u_0:

$$D_{\text{damp}}(u) = -\alpha(u - u_0),$$

with $\alpha > 0$.

Definition 8 (Geometry Coupling).

Let curvature be proportional to UBL:

$$R(\boldsymbol{x}, t) = \beta\, u(\boldsymbol{x}, t),$$

with $\beta > 0$. Define geometric feedback

$$F_{\text{geom}}(u) = \gamma R(u) = \gamma\beta u.$$

Definition 9 (Noise Term).

Let $\xi(\boldsymbol{x}, t)$be a stochastic field satisfying

$$\langle \xi(\boldsymbol{x}, t) \rangle = 0,$$
$$\langle \xi(\boldsymbol{x}, t)\xi(\boldsymbol{x}', t') \rangle = \eta(\boldsymbol{x}, t)\, \delta(\boldsymbol{x} - \boldsymbol{x}')\delta(t - t').$$

2. Lemmas

Lemma 1 (Variance Addition).

If $X_1, \ldots, X_N$ are independent with equal variance σ_0^2, then

$$\mathrm{Var}\left(\sum_{i=1}^{N} X_i\right) = N\sigma_0^2.$$

Proof. Independence implies

$$\mathrm{Var}(X + Y) = \mathrm{Var}(X) + \mathrm{Var}(Y).$$

Apply recursively.

Lemma 2 (UBL Noise Law).

Fractional noise in any independent system satisfies

$$\frac{\sigma}{\text{signal}} = \frac{1}{\sqrt{N}}.$$

Proof. From Lemma 1,

$$\sigma = \sqrt{N}\sigma_0.$$

Signal is proportional to N. Thus

$$\frac{\sigma}{N} = \frac{\sigma_0}{\sqrt{N}}.$$

Absorb σ_0 into units.

Lemma 3 (Planck Floor).

If $N = 1$, then

$$\frac{\sigma}{\text{signal}} = 1.$$

Proof. Set $N = 1$ in Lemma 2.

3. Theorems and Proofs

Theorem 1 (Substrate Evolution Equation).

The Unified Substrate Theory evolution equation for UBL is

$$\boxed{\frac{\partial u}{\partial t} = D\nabla^2 u - \alpha(u - u_0) + \gamma\beta u + \xi(\boldsymbol{x}, t)}$$

Proof. Combine:

1. **Diffusion** from tension:

$$\frac{\partial u}{\partial t}\,|_{\text{tension}} = D\nabla^2 u.$$

2. **Damping** (Definition 7):

$$-\alpha(u - u_0).$$

3. **Geometry feedback** (Definition 8):

$$+\gamma\beta u.$$

4. **Noise** (Definition 9):

$$+\xi(\boldsymbol{x}, t).$$

Sum all contributions.

Theorem 2 (Fixed Points of UST).

Spatially uniform fixed points satisfy

$$u^* = -\frac{\alpha u_0}{\gamma\beta - \alpha}.$$

Proof. Set $\nabla^2 u = 0$and $\xi = 0$. Then

$$\frac{du}{dt} = -\alpha(u - u_0) + \gamma\beta u.$$

Set derivative to zero:

$$0 = -\alpha u + \alpha u_0 + \gamma\beta u.$$

Solve for u:

$$u^* = -\frac{\alpha u_0}{\gamma\beta - \alpha}.$$

Theorem 3 (Planck Fixed Point).

There exist parameters $\alpha, \beta, \gamma, u_0$ such that

$$u^* = 1.$$

Proof. Set

$$1 = -\frac{\alpha u_0}{\gamma\beta - \alpha}.$$

Solve for any one parameter (e.g., u_0):

$$u_0 = -\frac{\gamma\beta - \alpha}{\alpha}.$$

Since parameters are free, choose values satisfying positivity constraints.

Theorem 4 (Cosmic Quiet Fixed Point).

There exist parameters such that

$$u^* = 10^{-123}.$$

Proof. Same as Theorem 3. Set

$$10^{-123} = -\frac{\alpha u_0}{\gamma\beta - \alpha}.$$

Solve for any parameter. Free parameters guarantee existence.

Theorem 5 (Classical Limit).

If $u \ll 1$ and $N \gg 1$, then UST reduces to classical field theory.

Proof. If $u \ll 1$, geometry feedback term $\gamma\beta u$ is negligible. If $N \gg 1$, noise term satisfies

$$\eta = \frac{1}{\sqrt{N}} \approx 0.$$

Thus

$$\frac{\partial u}{\partial t} \approx D\nabla^2 u - \alpha(u - u_0),$$

a classical diffusion-relaxation equation.

Theorem 6 (Quantum-Gravity Limit).

If $u \to 1$and $N \to 1$, then UST enters the Planck regime with order-unity fluctuations.

Proof. If $u \to 1$, curvature term $\gamma\beta u$becomes Planck-scale. If $N = 1$, Lemma 3 gives

$$\frac{\sigma}{\text{signal}} = 1.$$

Thus fluctuations are order unity, matching quantum-gravity behavior.

4. Final Result

The Unified Substrate Theory is formally defined by:

1. A substrate field ϕ.
2. A normalized activity field $u = \rho/\rho_P$.
3. A noise law derived from variance addition.
4. A tension term generating diffusion.
5. A damping term generating relaxation.
6. A geometry term coupling UBL to curvature.
7. A stochastic term encoding quantum randomness.
8. A single evolution equation unifying all of it.

$$\boxed{\frac{\partial u}{\partial t} = D\nabla^2 u - \alpha(u - u_0) + \gamma\beta u + \xi(\boldsymbol{x}, t)}$$

This equation has fixed points at:

- **UBL = 1** (Planck floor / black hole / bounce)
- **UBL = 10^{-123}** (our universe)

and reduces to classical physics when $u \ll 1$and $N \gg 1$.